Phase 2A Completion and Phase 2B Handoff

What has been built, what Candidate J proves, and how to start the next phase safely

> **Current status:** Phase 2A is complete. Candidate J passed the predeclared clean, stochastic pheromone-benefit gate on all three training seeds. The project can now plan Phase 2B, but attacks, datasets, detectors and robustness claims have not yet been created.

This guide is for a teammate beginning the next piece of work. It explains the result in plain language, points to the exact code and artifacts, and gives a disciplined Phase 2B starting sequence.

Project: Robust Stigmergic Coordination under Minority Pheromone Injection Attacks
Verified 1 November 2026 · Development-only simulator evidence

1. The project in one page

Agents work together in a simulated grid world. They search for food, pick it up, and deliver it back to a nest. They have only a small local view, but can leave and observe food and home pheromone fields. The main research question is whether a **local** defense can protect useful learned pheromone coordination from a minority of agents depositing misleading food pheromone.

Before attacking pheromone, we needed evidence that the learned policy actually benefits from it. That was the job of Phase 2A.

Phase	Question	Current state
Simulator and PPO foundation	Can the grid world, local observations and shared PPO policy run reproducibly?	Implemented and tested.
Phase 2A	Does a full-input policy do better than the same policy trained without pheromone?	**Passed by Candidate J** on the declared stochastic gate.
Phase 2B	Can bounded false-food pheromone degrade that useful behavior, and can local defenses help?	Ready for protocol planning; not yet run.

> **Do not overclaim:** Candidate J proves a clean development result, not attack robustness. It does not prove that every deterministic rollout coordinates well, that a detector works, or that a defense improves attacked performance.

2. What the agent can and cannot see

Each policy receives a 58-value local observation: a 3×3 patch with six values per cell, then four values describing its own state. The patch includes food, nest, food pheromone, home pheromone, other-agent occupancy and boundaries. The agent can also know whether it is carrying food, has picked up or delivered food, or was blocked.

The policy does **not** receive global coordinates, the global food map, agent identity, attacker identity, direct messages, privileged labels or a global-state critic. All policies use one parameter-shared PPO model with a 32×32 network and a shared delivery reward.

3. The fair Phase 2A comparison

For every seed, Phase 2A trains two separate models. This is essential: masking pheromone after training would not be a fair no-pheromone baseline.

Model	Input during both training and evaluation	Purpose
Full PPO	All 58 local values, including food and home pheromone.	The policy whose pheromone use we want to test.
No-pheromone PPO	The same 58-value shape, with exactly the 18 local food/home pheromone entries set to zero.	Tests whether pheromone adds value beyond food, nest, walls, nearby agents and own state.
Random action	No learned policy; legal actions sampled from a separate seeded RNG.	Checks that learned policies exceed chance behavior.

The phase used three PPO training seeds—7, 17 and 27—and diagnostic map seeds 100–103. The predeclared gate was strict: full PPO must have higher **stochastic mean delivered food** than both no-pheromone PPO and random action on every one of the three seeds.

4. Why earlier candidates did not pass

Candidate family	What was tested	What happened
A–C	Larger/repeated-retrieval eight-agent maps.	Mixed gains and losses across seeds.
D–E	Smaller maps with more repeated trips.	Closer results, but still at least one loss or tie.
F	Slower pheromone evaporation.	Trails became more visible, but full PPO still lost one seed.
G	Four agents, distant adjacent food cluster, 1,100-step horizon.	Won two seeds but tied at the 60-food maximum on seed 7.
H–I	Same scout-and-follow task with a 700-step deadline; then double training budget.	Full-input training became unstable; this direction was rejected.

These are useful negative results. Keep them; they justify why Candidate J changes only one thing.

5. What Candidate J does

Candidate J preserves Candidate G's stable scout-and-follow structure and increases only the food capacity. This removes the delivery ceiling that hid a speed advantage in Candidate G.

Setting	Candidate J value	Why it matters
Grid	12×12	Small enough for CPU PPO, but with a meaningful search route.
Agents	4	Creates a scout-and-follow opportunity without an eight-agent search swarm.
Food layout	Two adjacent cells far from the nest; location varies by reset seed.	One scout can discover a distant source; fixed-location memorization is discouraged.
Food capacity	50 units per cell; 100 total	The only change from G. It avoids the earlier 60-unit delivery ceiling.
Horizon	1,100 steps	Retains the stable long version rather than H/I's failed short deadline.
Field settings	Deposit 1.0, cap 10.0, evaporation 0.02	Retains the previously tested slow-fading field.
Training	300,000 requested agent transitions per policy	Same PPO settings, modes and seeds as Candidate G.

Candidate J assumptions

1. A full policy that can use local pheromone should turn the scout's discovered route into better team delivery than a separately trained no-pheromone policy.
2. Candidate G's seed-7 tie was partly caused by both policies reaching the old 60-unit maximum, not proof that pheromone had no value.
3. Increasing capacity while keeping the long horizon tests that explanation more cleanly than changing rewards, network size or several environment settings at once.
4. Seed-varying food locations prevent a policy from relying only on one memorized route.

A frozen-policy diagnostic supported the first assumption: masking only local pheromone changed Candidate G full-policy action distributions by total variation 0.272, 0.187 and 0.230 for seeds 7, 17 and 27, respectively. This shows input sensitivity, not causal proof by itself.

6. Candidate J result

All six PPO jobs and three paired random-action references completed. The table shows stochastic means, which are the predeclared advancement metric.

Training seed	Full PPO	No-pheromone PPO	Random	Full − no-pheromone	Gate outcome
7	100.00	68.75	3.00	+31.25	Pass
17	80.00	63.50	1.75	+16.50	Pass
27	100.00	80.50	2.25	+19.50	Pass

> **Conclusion:** Candidate J passed the three-seed stochastic development gate. It is now the clean learned-policy basis for Phase 2B.

There is an important limitation: deterministic diagnostics were mixed. Full PPO delivered 100.00, 0.00 and 60.00 deterministically for seeds 7, 17 and 27. Keep the Phase 2B clean and attacked evaluations stochastic unless a separately specified deterministic analysis is needed.

The runner also preserved an interrupted first invocation and resumed safely into fresh `-resume-1` paths. Nothing was overwritten. The final root manifest has six completed PPO rows, three random rows, three paired rows and a true gate.

7. Where the evidence is saved

- `artifacts/phase2a/scout-follow-capacity-run-001/paired.csv` — final same-seed comparisons.
- `artifacts/phase2a/scout-follow-capacity-run-001/runs.csv` — all PPO and random metrics.
- `artifacts/phase2a/scout-follow-capacity-run-001/manifest.json` — plan, provenance and resume record.
- `configs/phase2a/env/candidate_j_12x12_scout_follow_capacity.json` — Candidate J environment.
- `configs/phase2a/calibration-scout-follow-capacity.json` — declared Phase 2A matrix and gate.
- `docs/decisions/0012-phase2a-scout-follow-capacity.md` — decision, rationale and limitations.

8. How to start Phase 2B

Phase 2B should test whether a bounded minority false-food pheromone attack harms the useful Candidate J behavior, then whether a local defense helps. Do not jump directly to a large dataset: first freeze a small, auditable attack-calibration protocol.

> **Recommended first deliverable:** write a Phase 2B decision record before changing code or producing trajectories. It should name the Candidate J checkpoints, development maps, attacker fraction, injection budget, attack modes, clean/injection-disabled/attacked controls, metrics, and grouped train/validation/test split rules.

Phase 2B sequence

1. **Freeze the clean reference.** Use Candidate J's completed full PPO checkpoints unchanged. Do not retrain them after seeing attack results.
2. **Declare attack calibration.** Start with the existing bounded, nonnegative false-food injector. Deposits must occur only at selected attackers' actual occupied cells and use the same field cap and evaporation as honest deposits.
3. **Run three controls on the same maps.** Compare clean, injection-disabled and attacked conditions. Injection-disabled separates pheromone effects from simply losing productive workers.
4. **Measure attack effect first.** Report honest/team food delivered, completion, episode length, requested/authorized/applied injected mass, compromised-agent fraction and stochastic seed variation.
5. **Create grouped splits before windows.** Keep all variants from one complete map/episode group in the same split before generating temporal detector windows. Hold out maps and at least one attack configuration for final evaluation.
6. **Evaluate defenses fairly.** Compare undefended pheromone coordination, separately trained no-pheromone PPO, the local timeout baseline and a documented cautionary-pheromone adaptation. Defense inputs may use only local pheromone change, revisitation and the agent's own progress.
7. **Keep final claims separate.** Development calibration selects the protocol; held-out maps and attack settings support the final robustness claim.

> **Non-negotiable boundaries:** no pheromone erasure, negative deposits, global field writes, reward tampering, policy modification, attacker identities, global food locations, direct communication or privileged labels as defense inputs.

9. Useful commands and checks

Use the pinned Phase 2A environment from the repository root. Every output directory must be new.

Run the test suite

```
./.venv-phase2a/bin/python -m pytest -q
```

The verified post-Candidate-J result is **107 passed**.

Reproduce Candidate J in a new directory

```
./.venv-phase2a/bin/python -m stigmergy.cli phase2a-calibrate --plan configs/phase2a/calibration-scout-follow-capacity.json --output artifacts/phase2a/candidate-j-reproduction-001
```

If that exact new run is interrupted, resume only it with the same plan and output:

```
./.venv-phase2a/bin/python -m stigmergy.cli phase2a-calibrate --plan configs/phase2a/calibration-scout-follow-capacity.json --output artifacts/phase2a/candidate-j-reproduction-001 --resume
```

Read before implementing Phase 2B

- `AGENTS.md` — repository rules and research boundaries.
- `docs/CONTEXT.md` — current verified status.
- `docs/decisions/0012-phase2a-scout-follow-capacity.md` — Candidate J evidence.
- `src/stigmergy/attacks.py` and `src/stigmergy/trajectories.py` — existing bounded persistent-attack/provenance foundation.
- `src/stigmergy/defense.py` — local-history and timeout foundation.

10. Final handoff statement

We now have a clean, reproducible stochastic result showing that pheromone information helps the learned team in Candidate J. The correct next move is not to change Candidate J again. Freeze it, define Phase 2B's attack-calibration contract, then test whether bounded false pheromone can damage this useful coordination and whether strictly local defenses can recover it.

Prepared from the checked Candidate J artifacts, decision record and verified test suite. This document is a project handoff, not a final research result.